Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-based Seamless Indoor–Outdoor Positioning Technology in GNSS-challenged Environments			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	Vehicle Intelligence Technologies	Robotics/ Automation Machinery	Measurement/Sensor Technologies for Automation
Concept of Proposed Technology	Autonomous task execution by field robots deployed across diverse domains—such as construction sites, disaster safety response, defense and security missions, and logistics services—fundamentally depends on reliable, high-precision positioning technologies. Existing advanced localization technologies such as RTK-GNSS, UWB, inertial navigation, and SLAM-based positioning can provide high accuracy under specific conditions; however, they cannot independently ensure robust and seamless indoor–outdoor positioning continuity in field environments like construction and civil engineering sites, where spatial structures and work conditions continuously change. In such environments, frequent layout modifications, temporary structures, signal blockage, and intermittent sensor availability commonly occur. Conventional rule-based sensor switching and static			

	fusion approaches have inherent limitations in adaptability and predictive capability, making them insufficient to cope with the dynamic characteristics of field environments. To date, localization accuracy on the order of approximately 10 cm has been considered adequate for mobile robot navigation. However, high accuracy is required for advanced autonomous field robot operations such as remote manipulation in hazardous zones, automated smart construction tasks, and logistics operations replacing human workers. For example, global state-of-the-art Machine Control (MC) technology used in smart construction enables task execution at a precision level of 50 mm. Furthermore, autonomous field robot applications demand not only high horizontal positioning accuracy but also precise vertical (local Z-axis) positioning, and alignment with absolute elevation references is necessary to integrate with electronic maps, Building Information Model (BIM), and road/infrastructure design models. It is generally known that even with high-precision GNSS such as RTK-GPS, vertical accuracy is roughly twice as worse as the horizontal accuracy. Similarly, despite being recognized as the most accurate indoor positioning technology today, UWB-based localization typically achieves 7–25 cm horizontal (XY) error, while vertical (Z) error ranges from 20–90 cm, making it unsuitable for mission-critical field robot operations where both safety and operational precision are paramount. Therefore, this project aims to develop a seamless indoor–outdoor 3D positioning system that provides the ultra-high-precision localization required for autonomous field robot operations. By leveraging advanced sensor fusion, NLOS error mitigation, multimodal data, and AI deep learning models, the proposed system targets high performance levels in both indoor GNSS-challenged environments and outdoor open-sky environments, ultimately enabling highly reliable and accurate autonomous execution of field robotic tasks.
Rationale for Joint R&D Project	TNO, as an independent applied scientific research institute in Europe, possesses a full-cycle research structure spanning from basic science to application and industrial deployment. In particular, TNO demonstrates strong capabilities in the early-to-mid TRL stages (TRL 2–6) with excellence in fundamental scientific research, high-quality publications, and applied technology development. Korean SMEs, on the other hand, are highly optimized for prototype development, productization, and mass-manufacturing stages (TRL 6–9) with cost-efficient, high-quality production capabilities. Therefore, collaboration between the two organizations enables a complete end-to-end R&D value chain from concept creation to market deployment. Additionally, TNO has access to extensive testbeds as well as research and industrial networks across Europe, which would otherwise require high capital investment for Korean SMEs to establish independently. Through joint R&D, Korean companies can leverage these infrastructures while providing competitive manufacturing and rapid commercialization pathways, facilitating access to

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	European and global markets. TNO has significant experience with aligning developments and contributing to various standards. TNO will carry this expertise into this project, as a means of enabling successful overseas market expansion of Korean SMEs. This strategic collaboration allows both Korea and the Netherlands to competitively respond to dominant technology-leading nations, creating substantial technological, industrial, and market synergies.
Key Words	1. Seamless Indoor–Outdoor Positioning 2. AI-based Sensor Fusion 3. Centimeter-level 3D Localization 4. Autonomous Field Robotics 5. GNSS-challenged Environment Robustness

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	Develop a high-precision 3D **seamless indoor–outdoor positioning sensor system** applicable to autonomous field robot operations by sensor development and optimization as well as by leveraging **AI deep learning** technologies.			
R&D Target Specification	**Sensor fusion performance:** <20cm accuracy in x,y and z-directions for >99% of the time.			

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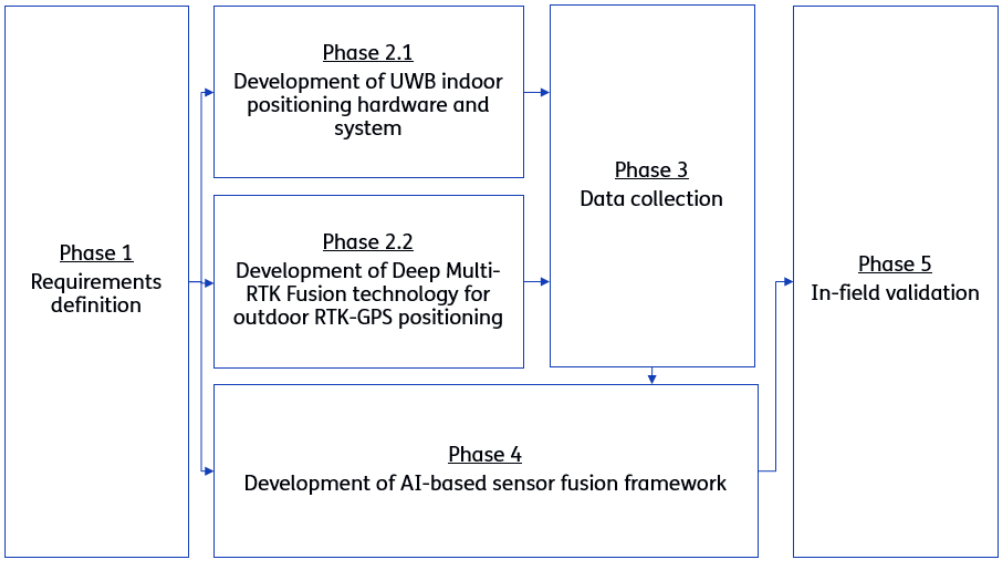
R&D Plan

TNO Proposal: AI-based Seamless Indoor–Outdoor Positioning Technology in GNSS-challenged Environments



Phase 1: Requirements definition (Months: 0-3; Lead: TNO)

- Analyze the use case scenario, and define high level target KPIs and specification for the UWB and RTK-GNSS systems, as well as the AI-based sensor fusion engine.

Milestone M1: Target specification sheet for the UWB and RTK-GNSS, and AI-based sensor fusion system.

Phase 2: Hardware development (Months: 0-12; Lead: SME)

2.1: Development of UWB indoor positioning hardware and system

- Combine the UWB Time Difference of Arrival (TDoA) and Angle of Arrival (AoA) data to reduce the Z-axis errors.
- Design a relocatable antenna array compatible with the target, suitable for civil and construction sites.
- Fusion with IMU and barometric sensors for optimized vertical accuracy.

2.2: Development of Deep Multi-RTK Fusion technology for outdoor RTK-GPS positioning

- Develop multi-band RTK-GPS hardware supporting automated driving applications.
- Achieve a target fix within 10 seconds
- Develop the RTK-GNSS software, including establishing connectivity to NTRIP correction servers in Europe/USA/Japan/Korea.

Milestone M2: UWB and RTK-GNSS systems developed and delivered to TNO, meeting the target specification defined in Phase 1.

Phase 3: Data collection (Months: 12-14; Lead: SME)

- Collect synchronized data from the developed sensor systems (and additional vehicle information) from a representative environment.
- Minimum duration of the data to be collected to be defined by TNO based on the AI model selected.
- Include ground truth information corresponding to sensor readings.

Milestone M3: Representative dataset of sensor data and ground truth collected and delivered to TNO.

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Phase 4: Development of AI-based sensor fusion framework (Months: 3-22; Lead: TNO)

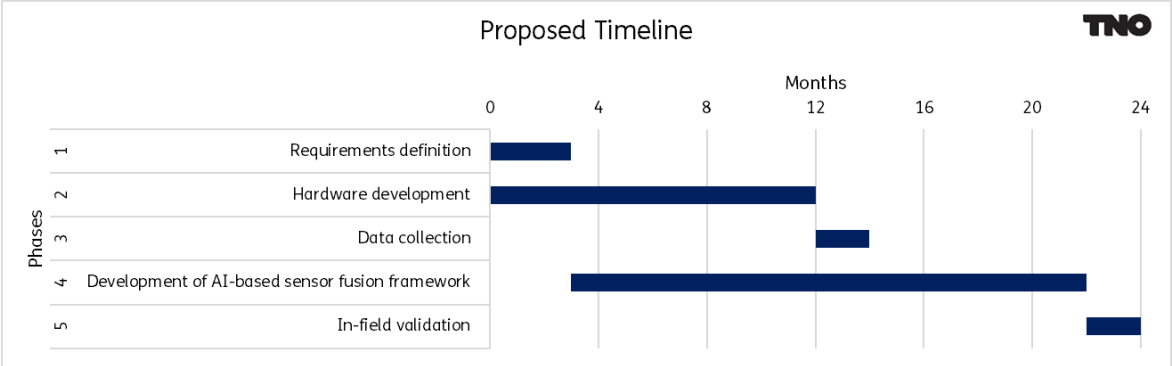
- Conduct research into applicable AI models and set up the pipeline. Relevant models will be analysed and a suitable model will be chosen.
- Format the data received from the SME such that it is ready for the training of the AI models.
- Development/training AI models based on the collected data.
- Perform first validation on a test set of the data (software based).
- Perform a short validation study on the TNO Carlabs, particularly covering the transition scenarios (indoor-outdoor).

Milestone M4: AI-based sensor fusion framework developed with the target of achieving the R&D specifications defined above.

Phase 5: In-field validation (Months: 22-24; Lead: SME)

- Design a controlled experiment involving relevant equipment operating in a representative environment.
- Conduct in-field validation including collecting ground truth data.
- Analyse and compare localization metrics achieved by the developed sensor modules; AI-based sensor fusion module and ground truth measurements.

Milestone M5: Report on analysed data and performance metrics.



4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• TNO will lead the development of an AI-based sensor fusion engine that combines the outputs of the various sensors. • Requirements definition based on the scenarios. • This is trained on (amongst other sensors) the indoor UWB and outdoor RTK-GNSS, developed by the partner SME. • The fusion framework also provides a refined Z-axis position estimate using multimodal sensor information, including the Z-axis output of the RTK-GNSS, UWB, IMU and barometer. • Special emphasis is paid to the mitigation of NLOS and multipath using relevant AI-models. • Conduct validation testing on a test dataset (software environment), and using the TNO Carlabs. • Support the SME in design of test protocols, and defining validation plans.

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Expected R&R of Korean SME Partner	• Korean SME will lead hardware/system implementation, field validation. • Develop UWB indoor positioning hardware/system optimized for construction/civil engineering sites with relocatable anchor configurations. • Develop RTK-GNSS hardware and software supporting multi-band/multi-GNSS operation. • Integrate multimodal sensors (IMU, barometer, LTE/5G communication modules) and embedded systems • Manufacture prototype positioning sensor units for autonomous field robot applications • Conduct field validation and demonstration tests in real GNSS-challenged construction environments

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Achieve centimeter-scale 3D seamless indoor–outdoor positioning performance applicable to autonomous field robot operations in GNSS-challenged environments • Secure vertical precision technology (local Z-axis) optimized for construction/civil infrastructure, enabling alignment with BIM, digital maps, and elevation design data • Develop AI-based adaptive sensor fusion algorithms that improve robustness against environmental variability such as NLOS and multipath conditions • Advance TRL from TRL 2 → TRL 5 enabling field testing on the path towards commercialization in global positioning markets • Enable safer and more efficient autonomous operations in hazardous construction and industrial sites by replacing or supporting human labor • Establish a positioning platform that can be extended to industrial logistics, defense/security robots, disaster response, and next-generation smart construction

6. References

(1) AI-enabled Enhanced Object Classification In Dynamic Environments (CIDER); M. Braat, R. Smit; TNO Report, 2022.

(2) SDS++: Online Situation-Aware Drivable Space Estimation for Automated Driving; M.M. Sanchez, G. Trots, R. Smit, P.V. Oliveira, J. Elfring, R. v.d. Molengraft, E. Silvas; IEEE Transactions on Intelligent Vehicles, 10 (3) p.p. 2140-2151.